Next-generation Bayesian local robustness

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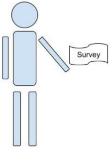
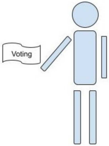
The basic problem

We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.





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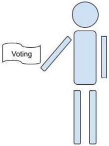

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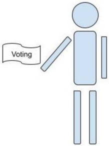

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This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}\left[y | x, \theta\right] = m(\theta^\intercal x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta | \text{Survey data})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}}) = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta | \text{Survey data})}\left[y | x_j\right]$.

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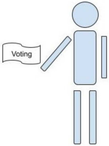

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Problem: MCMC is time consuming and the regression model is surely misspecified.
Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

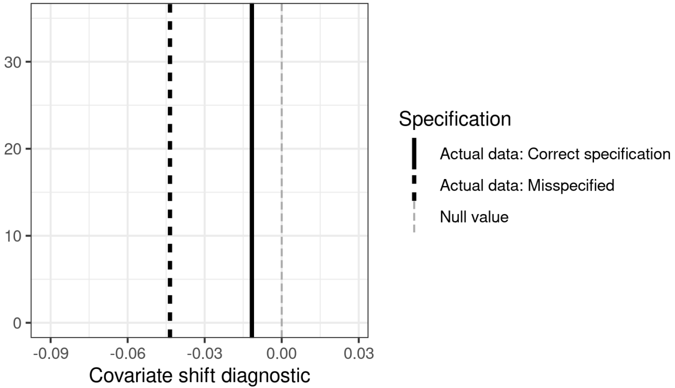

Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecied model is more extreme.

Problem: What is "large enough" to be real evidence of misspecification?
Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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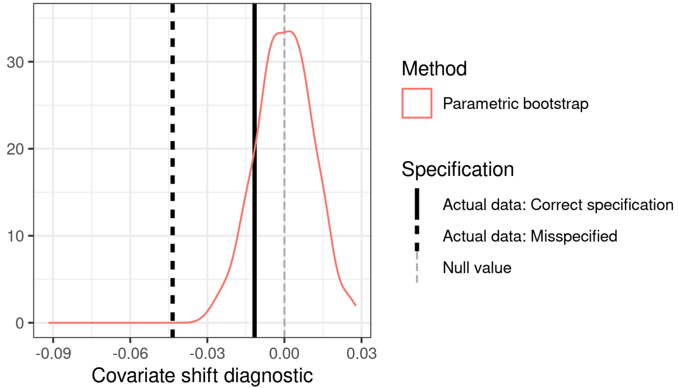

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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.
Possible answer: Infintesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

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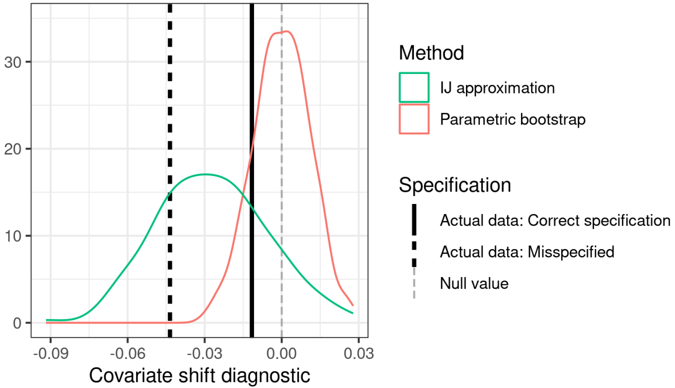

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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

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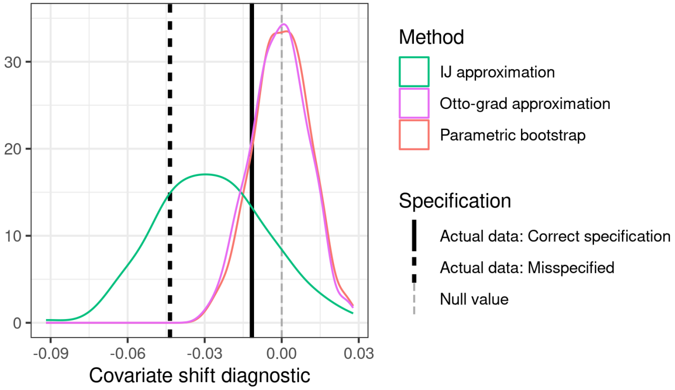

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Answer: Our flow-based local sensitivity measures ("Otto grad") produce good approximations to the bootstrap at negligible computational expense.

Outline

- Bayesian local regression specification checks
  - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
  - (With Lucas Schwengber)

Bayesian local regression specification checks

Local specification checks

$$\hat{\mu}^{\mathrm{MrP}}(Y_{\mathcal{S}}) = \frac{1}{N_T}\sum_{j=1}^{N_T}\hat{y}_j \text{ where } \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[m(\theta^\intercal x_j)\right]}_{\text{Want to check without re–running MCMC}}.$$

This model of *survey data* is definitely misspecified.

Why do we care about correct specification? We want to use $\mathcal{P}_S(x)$ to learn about $\mathcal{P}_T(x)$.

Correct specification $\Rightarrow$ Regression estimates are invariant to $\mathcal{P}_S(x)$

Idea: Check invariance to $\mathcal{P}_S(x)$ directly.

Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y.
Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x)\mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x))\,\mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial\mu(\mathcal{P}_\epsilon)/\partial\epsilon$
5. Plug the empirical distribution into $\Delta(w, \hat{\mathcal{P}}_S)$ and check whether ≈ 0

Bayesian posterior functionals

We are interested in $\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[g(\theta)\right]$ where $g(\theta) := \frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j)$.
Let the regression log likelihood be $\ell(y|x,\theta)$.
Define the "generalized posterior" functional:

$$T(\mathcal{P}, N) := \frac{\int g(\theta)\exp\left(N\int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right)\pi(\theta)d\theta}{\int \exp\left(N\int \ell(y|x,\theta)\mathcal{P}(x,y)dxdy\right)\pi(\theta)d\theta}.$$

Let $\mathbb{F}_N$ denote the empirical distribution over x_i, y_i. Then

$$\mathbb{E}_{\mathcal{P}(\theta|\text{Survey data})}\left[g(\theta)\right] = \frac{\int g(\theta)\exp\left(N\frac{1}{N}\sum_{n=1}^N \ell(y_i|x_i,\theta)\right)\pi(\theta)d\theta}{\int \exp\left(N\frac{1}{N}\sum_{n=1}^N \ell(y_i|x_i,\theta)\right)\pi(\theta)d\theta} = T(\hat{\mathcal{P}}_S, N).$$

The directional derivative is

$$\left.\frac{\partial T(\mathcal{P}_\epsilon, N)}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{P},N)}\left(g(\theta), N\int \ell(y|x,\theta)w(dx)\mathcal{P}(y|x)dxdy\right)$$

Our diagnostic is then given by

$$\Delta(w, \hat{\mathcal{P}}_S) = \mathrm{Cov}_{\mathcal{P}(\theta|\text{Survey data})}\left(\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j), \sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta)\right)$$

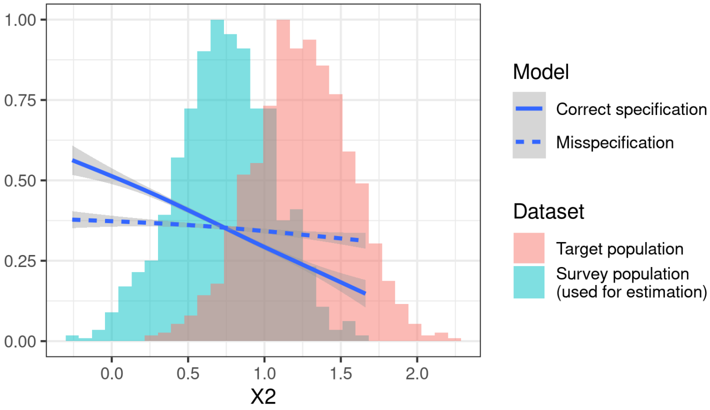


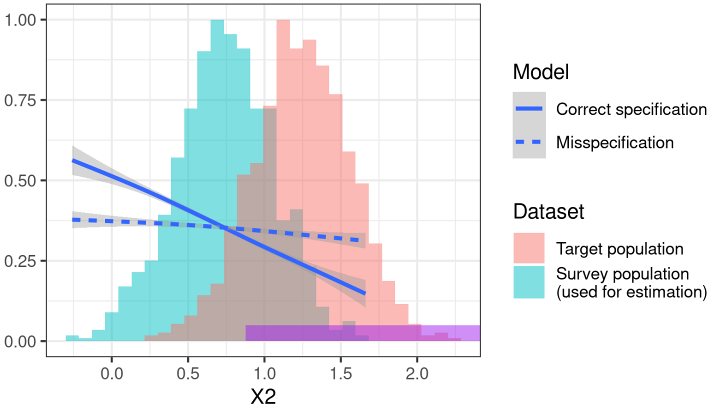
Figure 2: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

$$\text{Model 0 (correct)} : \mathtt{y} \sim \text{Logistic}(\mathtt{1 + X1 + X2 + X3 + X4 + X5})$$

$$\text{Model 1 (misspecified)} : \mathtt{y} \sim \text{Logistic}(\mathtt{1 + X3 + X4 + X5})$$

Comput diagnostic with $w(x) = \mathbb{I}\left(\mathtt{X2} > 0.878\right)$



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References

R. Giordano and T. Broderick. The Bayesian infinitesimal jackknife for variance. *arXiv preprint arXiv:2305.06466*, 2026.